Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/06/2017
PI Name:	Barstow, Thomas Ph.D.		
Project Title:	Determining the Effect of Space Flight on the Incidence of Cardiovascular Risk Factors and Disease		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	66506-0109	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	05/01/2015	**End Date:**	08/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/31/2017 per F. Hernandez/ARC and NSSC information (Ed., 1/24/17) NOTE: Extended to 10/31/2016 per F. Hernandez/ARC (Ed., 5/13/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ade, Carl Ph.D. (Kansas State University)		
Grant/Contract No.:	NNX15AK01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	CONTINUATION OF "Determining the Effect of Space Flight on the Incidence of Cardiovascular Risk Factors and Disease," GRANT NNX13AQ93G. Cardiovascular disease (CVD) is a primary cause of morbidity and mortality among the general population. Early detection and identification provides the foundation for mitigating the disease progression. Traditional risk stratification models, such as the Framingham Risk Score, are used to classify an individual as low, moderate, or high-risk. While this model is useful, CVD prediction can be improved with additional clinical measurements, specifically in special populations like military and commercial airline pilots, thus providing evidence that a generalized CVD prediction model may not be adequate for individuals such as astronauts. More clearly defining the CVD risk factors associated with spaceflight is critical for future long-duration missions. The astronaut is challenged with an environment that has minimal medical capabilities and treatment options relative to Earth-based occupations. Likewise, mission success and survival depends upon the health and physical capabilities of its crew. A cardiac event, like a myocardial infarction or chronic arrhythmia, will limit an astronaut's ability to perform in space and may require mission termination. In addition, the prevention of CVD and increased understanding of the relationship between space flight and CVD progression are of critical importance. Therefore the overall goal of the current proposal is to use data from the LSAH (Longitudinal Surveillance of Astronaut Health) and LSDA (Life Sciences Data Archive) to determine whether spaceflight induces a higher incidence of cardiovascular risk factors and disease in astronauts. The deliverables from this project will go beyond those specified as Human Research Program (HRP)-Gaps and include determining the rate of progression for cardiovascular disorders and evaluating the effects of mission duration and radiation exposure on the incidence of a cardiovascular event, disorder, and risk factor. We believe this project will not only provide relevant data for immediate use, but also a clinical model that will be invaluable for the future of exploration missions. Redirected Primary Aims for Continuation NNX15AK01G: The below redirected aims were derived from formal and informal conversations with the cardiovascular research team at NASA-Johnson Space Center, LSAH scientists/staff, and scientists within general cardiology, cardiovascular epidemiology, and microgravity physiology. It is the consensus of these parties that given the complexity of the project's overriding goal to determine whether spaceflight induces a higher incidence of cardiovascular risk factors and disease in astronauts, additional comparison populations are required. Likewise, matching groups by a single variable, like age, excludes several covariates (e.g., fitness, LDL, body mass index) which may misrepresent the true underlying prevalence of CVD in astronauts compared to the general population. Therefore, we believe these additional aims more comprehensively address the IRP (Integrated Research Plan) GAP CV8: Can manifestations of sub-clinical or environmentally induced cardiovascular diseases during spaceflight be predicted? 1. Quantify the prevalence of "hard cardiovascular events" in astronauts compared to controls when matched for age class only. Matched controls will be drawn from the Dallas Heart Study, and/or Cooper Clinic database, and LSAH. 2. Quantify the prevalence of "hard cardiovascular events" in astronauts compared to controls when matched for age class and CVD risk profile (e.g., coronary artery calcium score, body mass index, Framingham risk per 10 years, and smoking). Matched controls will be drawn from the Dallas Heart Study, and/or Cooper Clinic database, and LSAH. 3. Quantify the prevalence of "hard cardiovascular events" in astronauts compared to controls when matched for age class and fitness. Matched controls will be drawn from the Dallas Heart Study, and/or Cooper Clinic database, and LSAH.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The investigators acknowledge the limitations imposed with this type of retrospective analysis. However, given the challenges and time commitment required to rigorously collect this type of data in astronauts immediately pre- and post-flight this study design will provide a relatively rapid and more cost effective analysis that can be used to guide future, more highly controlled studies. In addition, given the present proposal's retrospective analysis, the findings can be directly compared to the longitudinal data on the incidence of cardiovascular disease previously evaluated by the current investigators in this same cohort of astronauts.
Task Progress:	What is new? • The results of this study provide insight into the cardiovascular disease risk (CVD) in a large cohort of NASA astronauts. • Career NASA astronauts as a whole had the same risk of developing a clinical CVD endpoint or a specific CAD endpoint as did a group of NASA civil servant employees, and this risk did not change after adjusting for known CVD covariates. • Among the NASA astronauts who had had spaceflight experience by the 5th decade of life, the risk of CVD was not different from that of astronauts without a history of spaceflight by the 5th decade of life. What are the clinical implications? • These findings have important implications for the lifelong assessment of astronaut health and suggest that individuals in the astronaut corps are not at an increased risk of developing CVD and that spaceflight experience may not contribute to the risk of CVD events. • The broadly defined spaceflight experiences do not appear to put NASA astronauts at greater risk of critical CVD endpoints across the lifespan. • Whether specific types of missions like long-duration missions or those with greater exposure to cosmic radiation beyond low-earth orbit elicit long-term CVD risk merits further study.
Bibliography Type:	Description: (Last Updated: 01/23/2020)
Articles in Peer-reviewed Journals	Ade CJ, Broxterman RM, Moore AD, Barstow TJ. "Decreases in maximal oxygen uptake following long-duration spaceflight: Role of convective and diffusive O(2) transport mechanisms." J Appl Physiol (1985). 2017 Apr 1;122(4):968-75. Epub 2017 Feb 2. https://doi.org/10.1152/japplphysiol.00280.2016 ; PubMed PMID: 28153941 , Apr-2017
Articles in Peer-reviewed Journals	Ade CJ, Broxterman RM, Charvat JM, Barstow TJ. "Incidence rate of cardiovascular disease end points in the National Aeronautics and Space Administration Astronaut corps." J Am Heart Assoc. 2017 Aug 7;6(8):e005564. https://doi.org/10.1161/JAHA.117.005564 ; PubMed PMID: 28784652; PubMed Central PMCID: PMC5586420 , Aug-2017